

SOC HOME LAB

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EXECUTIVE SUMMARY

This project presents the design and implementation of a virtualized Security Operations Center (SOC) Home Laboratory for monitoring, analyzing, and investigating security events within a controlled cybersecurity environment.

The laboratory was designed to simulate the basic workflow of an enterprise SOC by combining pfSense Firewall, Splunk Enterprise, Windows Server 2019, Kali Linux, NetFlow/softflowd, and VMware Workstation. The environment provides both network-level and endpoint-level visibility, allowing security events to be generated, collected, monitored, and investigated.

The pfSense firewall serves as the central network security gateway. It provides network segmentation, traffic filtering, firewall logging, and network-flow monitoring. Firewall traffic and NetFlow data are collected to provide visibility into communication between internal systems and external destinations.

Windows Server 2019 functions as the primary Windows-based monitored system and generates security and authentication events. These events are forwarded to Splunk Enterprise, which acts as the centralized SIEM platform for log collection, searching, visualization, and security analysis.

Kali Linux is used as the controlled security-testing system. Security testing activities, including authentication testing and network traffic generation, are performed against the laboratory systems to produce realistic security telemetry. The resulting events are monitored through pfSense and Splunk.

The project demonstrates an end-to-end security monitoring workflow:

Security Activity → Network/Endpoint Logs → Collection → Splunk SIEM → Detection → Investigation

The implementation provides practical exposure to firewall administration, network monitoring, SIEM operations, Windows security-event analysis, controlled attack simulation, and security investigation. The laboratory therefore serves as a practical environment for understanding how multiple security technologies operate together within a SOC.

All testing activities documented in this report were conducted within the controlled laboratory environment for educational and cybersecurity training purposes.

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1. INTRODUCTION

1.1 Project Overview

The SOC Home Lab was designed using pfSense, Splunk Enterprise, Windows Server 2019, Kali Linux, NetFlow/softflowd, and VMware Workstation. Each component has a defined role within the security architecture.

pfSense operates as the network security gateway and provides firewall filtering, network traffic logging, and flow monitoring. Windows Server generates system and authentication security events that can be collected for analysis. Splunk Enterprise functions as the centralized SIEM platform where security telemetry can be searched, correlated, and investigated. Kali Linux is used to perform controlled security-testing activities that generate realistic security events within the laboratory.

The overall implementation demonstrates how network and endpoint telemetry can be brought together into a centralized monitoring environment.

1.2 Project Objectives

The primary objectives of the project are:

- To design and deploy a functional virtualized SOC environment.
- To implement pfSense as a network firewall and security gateway.
- To monitor network traffic and firewall activity.
- To implement NetFlow-based network visibility using softflowd.
- To collect Windows security and authentication events.
- To configure Splunk as a centralized SIEM platform.
- To generate controlled security-testing activity within the laboratory.
- To identify and analyze security-relevant events.
- To correlate network and endpoint telemetry during investigation.
- To develop practical skills in SOC monitoring and security analysis.

1.3 Scope

The project is limited to a controlled virtual cybersecurity laboratory. The implementation covers firewall configuration, network segmentation, traffic monitoring, firewall logging, NetFlow collection, Windows security-event monitoring, centralized SIEM analysis, and controlled security testing.

All security-testing activities documented in this report are performed against systems within the laboratory environment for educational and cybersecurity training purposes.

1.4 Technologies Used

The major technologies used in the implementation are:

Technology	Primary Role
pfSense	Firewall, gateway, traffic filtering and network logging
Splunk Enterprise	SIEM, centralized log collection and analysis
Windows Server 2019	Windows infrastructure and security-event source
Kali Linux	Controlled security-testing platform
NetFlow / softflowd	Network-flow monitoring
VMware Workstation	Virtualization platform

The following chapters document the implementation of the individual components and demonstrate how they operate together to provide an integrated SOC monitoring and investigation environment.

2. SOC HOME LAB ARCHITECTURE

2.1 Architecture Overview

The SOC Home Lab was designed as a virtualized cybersecurity environment in which network infrastructure, endpoint systems, security monitoring, and controlled security-testing components operate together within an isolated laboratory network.

The architecture is centered around pfSense, which functions as the primary network security gateway between the external network and the internal laboratory segments. Network traffic generated by the internal systems passes through pfSense, allowing traffic to be filtered, logged, and monitored.

The environment also includes a Windows Server system used for Windows-based infrastructure and security-event generation, a Kali Linux system used for controlled security testing, and Splunk Enterprise used as the centralized SIEM platform. Network-flow information is collected using softflowd/NetFlow, while endpoint and security logs are forwarded to Splunk for centralized analysis.

The architecture follows the general security-monitoring flow:

Network / Endpoint Activity → Security Controls → Log & Flow Collection → Splunk SIEM → Analysis & Investigation

2.2 Laboratory Components

The major components of the laboratory architecture are described below.

pfSense Firewall

pfSense is positioned at the network boundary and acts as the primary firewall and gateway. It provides:

- Network routing
- Stateful firewall filtering
- Network segmentation
- Firewall traffic logging
- DNS-related filtering
- pfBlockerNG-based filtering
- Network-flow monitoring
- Visibility into internal and external communications

The configured interfaces provide connectivity between the external/WAN network and the internal laboratory networks.

Windows Server

Windows Server functions as an important infrastructure and telemetry source within the laboratory. It generates Windows system and security events that can be collected and analyzed through the SIEM platform.

Security-relevant activities include authentication events, failed authentication attempts, system activity, and other Windows security events generated during normal operation and controlled testing.

Kali Linux

Kali Linux is used as the controlled security-testing platform. It provides the tools required to generate security-testing activity against authorized laboratory systems.

Testing activity is intentionally performed within the isolated environment so that the resulting network and endpoint events can be observed without targeting external systems.

Splunk Enterprise

Splunk Enterprise functions as the centralized SIEM platform. Security telemetry generated by the laboratory components is collected and made available for searching and analysis.

Splunk provides:

- Centralized log collection
- Event searching
- Security-event analysis
- Log correlation
- Source-based investigation
- Detection development
- Visualization and monitoring

NetFlow / softflowd

NetFlow-based monitoring provides flow-level visibility into network communications. softflowd operates on the pfSense environment and records network-flow information that can be used to analyze communication patterns, protocols, source addresses, destination addresses, and traffic behavior.

VMware Workstation

VMware Workstation provides the virtualization layer on which the laboratory systems operate. This allows multiple operating systems and security appliances to run independently while maintaining controlled virtual networking between them.

2.3 Network Segmentation

The laboratory uses multiple network segments to separate different traffic paths.

The pfSense configuration contains the following interfaces:

Interface	Address	Function
WAN	192.168.241.132/24	External/WAN connectivity
LAN	192.168.10.1/24	Primary internal laboratory network
OPT1	192.168.20.10/24	Additional internal laboratory segment

This segmentation allows traffic between different laboratory components to pass through a controlled network-security boundary rather than placing every system on a single unrestricted network.

The use of separate segments also improves the ability to identify the source and direction of network activity during investigation.

2.4 Security Data Flow

The architecture was designed to allow security activity to be observed at multiple layers.

When a system generates network traffic, the traffic is processed by pfSense. The firewall evaluates the traffic against its configured policies and records relevant activity in its firewall logs.

At the network-flow level, softflowd provides summarized flow information. This enables analysis of network communication without depending exclusively on raw packet capture.

Windows systems generate endpoint and authentication events that provide host-level visibility. These events complement the network-level information obtained from pfSense.

The collected telemetry is then made available to Splunk, where it can be searched and analyzed as part of the SOC monitoring workflow.

This creates multiple sources of security evidence:

Network Layer: pfSense firewall and NetFlow data

Endpoint Layer: Windows security and system events

Testing Layer: Kali-generated controlled security activity

SIEM Layer: Splunk centralized analysis

2.5 SOC Monitoring Workflow

The implemented environment follows a structured monitoring workflow:

1. Activity Generation
Normal system activity and controlled security-testing activity generate network and endpoint events.
2. Network Enforcement
pfSense processes network traffic and applies the configured firewall and filtering policies.
3. Telemetry Generation
Firewall logs, network-flow records, and Windows security events provide security telemetry.
4. Centralized Collection
Relevant logs are forwarded or collected for centralized analysis.
5. SIEM Processing
Splunk provides a centralized location for searching and examining the collected events.
6. Security Analysis
Events are examined to identify authentication failures, unusual network behavior, and other security-relevant activity.
7. Investigation
Related events can be correlated across network and endpoint sources to understand the activity and determine its significance.

2.6 Architecture Diagram

The architecture diagram illustrates the relationship between the major components of the SOC Home Lab. pfSense is positioned as the central network security device, while Windows Server and Kali Linux operate within the controlled laboratory environment. Network and endpoint telemetry is subsequently analyzed through Splunk.

SOC Home Lab Architecture

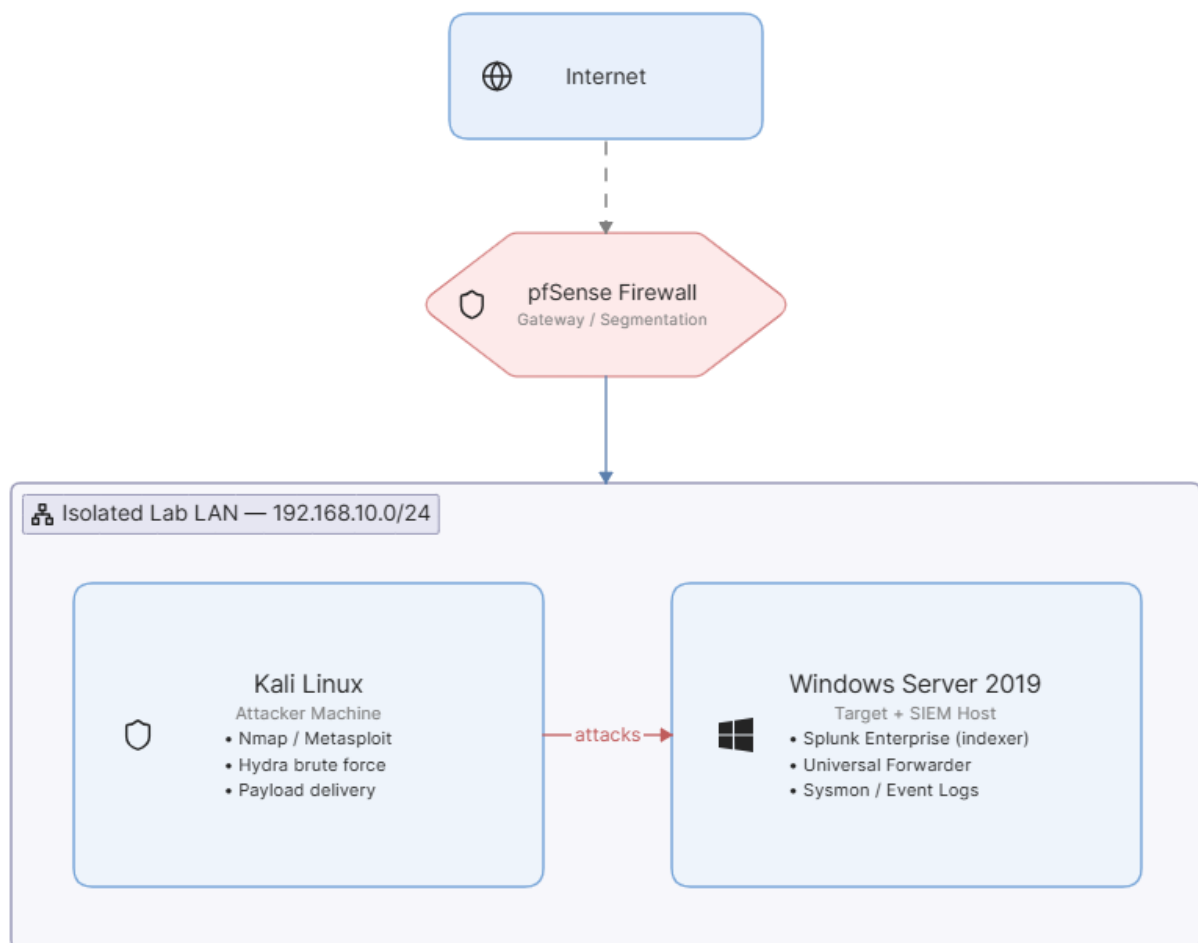


Figure 1: SOC Home Lab Network and Security Monitoring Architecture

The architecture establishes a clear separation between security enforcement, activity generation, telemetry collection, and security analysis. This separation is important because each component has a specific responsibility rather than duplicating the function of another component.

2.7 Security Monitoring Objective

The primary purpose of this architecture is to provide visibility across both network and endpoint activity.

Rather than relying on a single source of information, the laboratory combines firewall logs, network-flow data, and Windows security events. This allows an analyst to examine an event from multiple perspectives.

For example, a controlled authentication attack may generate Windows authentication failures at the endpoint while simultaneously producing network traffic that can be observed through pfSense. Splunk provides the centralized location where these different sources can be searched and correlated.

The resulting architecture therefore demonstrates the fundamental SOC concept of centralized security visibility across multiple telemetry sources.

3. VIRTUALIZED INFRASTRUCTURE

3.1 Virtualization Environment

The SOC Home Lab was implemented as a virtualized environment using VMware Workstation. Virtualization was selected to allow multiple operating systems and security platforms to operate simultaneously on a single physical host while maintaining logical separation between laboratory components.

The virtualized approach provides an isolated and reproducible environment for cybersecurity experimentation. It also allows individual systems to be configured, tested, restarted, or modified without affecting the host operating system or external production systems.

The laboratory consists of dedicated virtual machines representing the network security gateway, endpoint/infrastructure systems, SIEM platform, and security-testing workstation.

3.2 Virtual Machines

The primary virtual machines used within the laboratory are:

Virtual Machine	Operating System / Platform	Primary Function
Firewall VM	pfSense	Network security gateway and firewall
Server VM	Windows Server	Infrastructure and security-event source
Client / Testing VM	Kali Linux	Controlled security testing
SIEM	Splunk Enterprise	Centralized security monitoring and analysis

Each virtual machine has a separate operational role within the SOC architecture. This separation prevents the security functions of one component from being unnecessarily mixed with another.

3.3 VMware Network Configuration

VMware virtual networking provides the communication paths between the laboratory systems. The virtual network adapters are assigned according to the role of each system within the SOC architecture.

The pfSense virtual machine contains multiple network interfaces to support WAN and internal network connectivity. Internal systems communicate through the appropriate pfSense interface rather than directly bypassing the firewall.

This design ensures that internal traffic can be subjected to firewall policies and can generate network telemetry for subsequent analysis.

3.4 Resource Allocation

The virtual machines were allocated system resources according to their workload requirements. The resource allocation was designed to balance the requirements of the SOC components with the limitations of the physical host system.

The laboratory uses lightweight configurations where possible because the environment is intended for security monitoring and testing rather than production workloads.

Resource allocation includes:

- Virtual CPU allocation
- Virtual memory allocation
- Virtual storage
- Virtual network adapters
- Network connectivity configuration

The exact resource allocation may vary during testing depending on the workload generated by Splunk, Windows Server, pfSense, and the security-testing environment.

3.5 Isolation and Security Considerations

The virtualized environment provides an important security boundary for controlled testing. Security-testing tools running on Kali Linux are directed toward authorized laboratory systems rather than external infrastructure.

Network segmentation through the pfSense virtual firewall further restricts direct communication between different laboratory segments.

This approach provides several advantages:

- Controlled attack-testing environment
- Reduced risk to external systems
- Reproducible security experiments
- Independent operating-system environments
- Centralized network control
- Easier troubleshooting and recovery
- Ability to modify configurations without affecting physical infrastructure

3.6 Role of Virtualization in the SOC Lab

Virtualization is a fundamental part of the laboratory because the SOC requires multiple systems to generate and process different types of security telemetry.

The environment can therefore be viewed as four major functional layers:

Infrastructure Layer

Provides the virtual machines and networking required for the laboratory.

Security Enforcement Layer

pfSense controls and monitors network traffic.

Telemetry Layer

Windows security events, firewall logs, and network-flow information provide security data.

Analysis Layer

Splunk centralizes and analyzes the collected security telemetry.

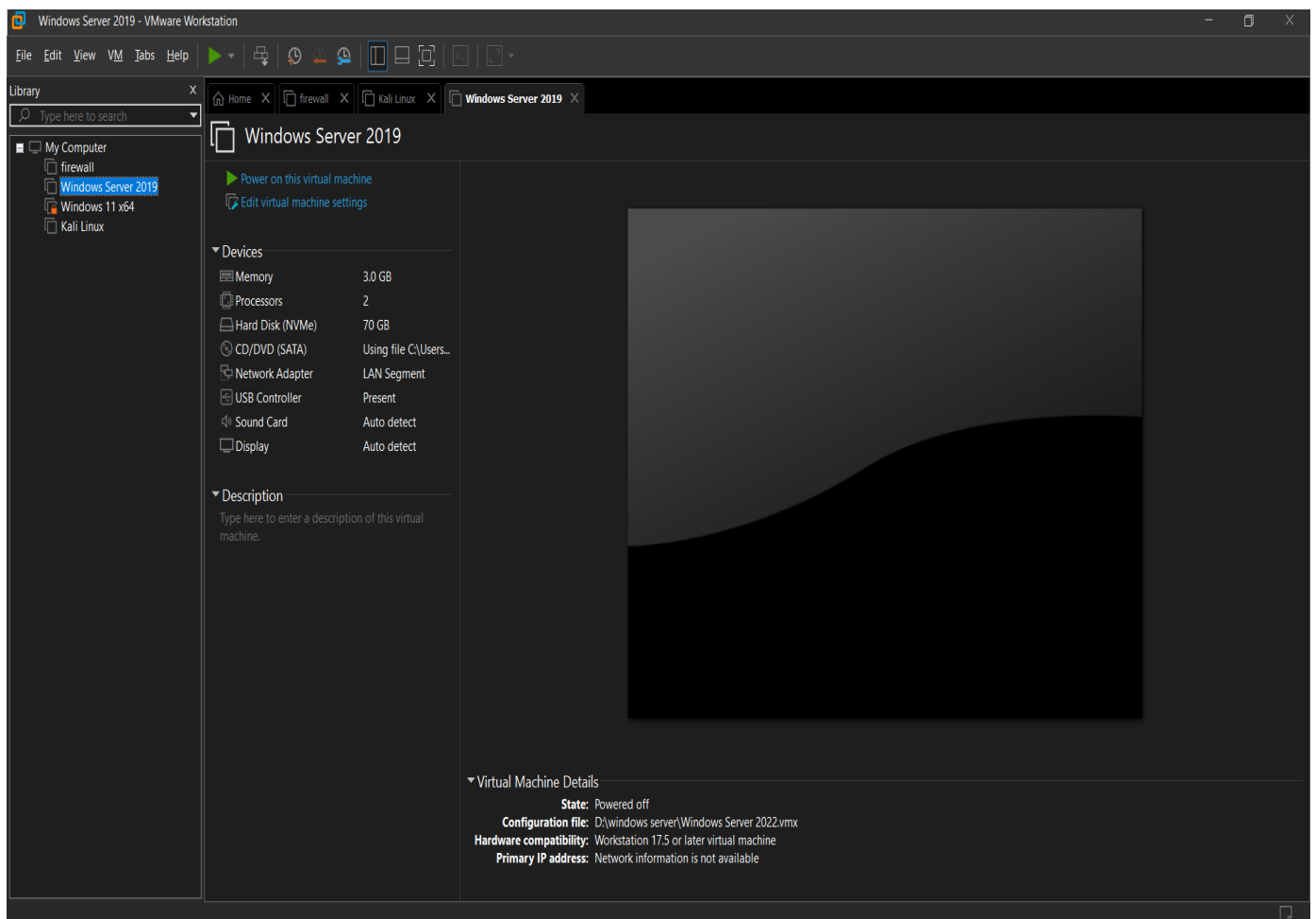


Figure 2: VMware-Based SOC Home Lab Virtual Infrastructure

This layered design allows individual components to be tested independently while still demonstrating how they operate together as an integrated SOC environment.

4. PFSense FIREWALL & NETWORK SECURITY

4.1 pfSense Deployment

pfSense Community Edition was deployed as a virtual firewall within the VMware Workstation environment. The firewall serves as the primary network-security gateway of the SOC Home Lab and provides centralized control over traffic entering, leaving, and moving between the laboratory network segments.

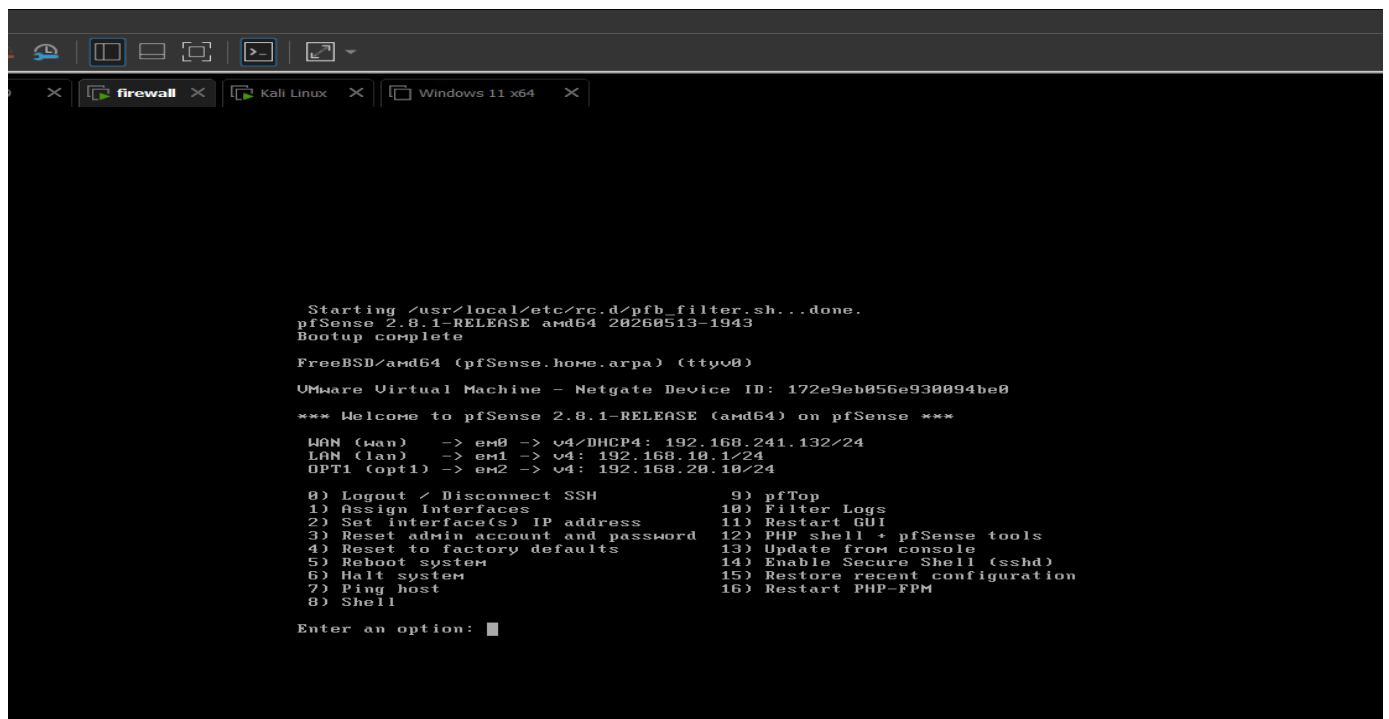
The pfSense virtual machine was successfully initialized and its network interfaces were detected and assigned during the deployment process. The console interface provides direct access to the firewall system and confirms the operational status of the deployed pfSense instance.

The deployed system contains three network interfaces:

- **WAN — em0:** 192.168.241.132/24
- **LAN — em1:** 192.168.10.1/24
- **OPT1 — em2:** 192.168.20.10/24

The WAN interface provides connectivity toward the external network, while the LAN and OPT1 interfaces are used for internal laboratory network connectivity. This multi-interface configuration enables pfSense to operate as the central routing and security boundary for the virtual SOC environment.

The deployment establishes the foundation for the subsequent security controls

A screenshot of a terminal window titled 'firewall' showing the pfSense boot process. The output includes the version '2.8.1-RELEASE amd64', the hostname 'pfSense.home.arpa', and the VMware Virtual Machine ID. It then displays the assigned IP addresses for the WAN (em0), LAN (em1), and OPT1 (em2) interfaces. A menu of 16 options is shown, including logging out, assigning interfaces, setting IP addresses, resetting accounts, rebooting, and enabling SSH. The prompt 'Enter an option:' is at the bottom.

```
Starting /usr/local/etc/rc.d/pfb_filter.sh...done.
pfSense 2.8.1-RELEASE amd64 20260513-1943
Bootup complete

FreeBSD/amd64 (pfSense.home.arpa) (ttyv0)
VMware Virtual Machine - Netgate Device ID: 172e9eb056e930094be0

*** Welcome to pfSense 2.8.1-RELEASE (amd64) on pfSense ***

WAN (wan)  -> em0 -> v4/DHCP4: 192.168.241.132/24
LAN (lan)   -> em1 -> v4: 192.168.10.1/24
OPT1 (opt1) -> em2 -> v4: 192.168.20.10/24

0) Logout / Disconnect SSH          9) pfTop
1) Assign Interfaces                10) Filter Logs
2) Set interface(s) IP address     11) Restart GUI
3) Reset admin account and password 12) PHP shell + pfSense tools
4) Reset to factory defaults        13) Update from console
5) Reboot system                    14) Enable Secure Shell (sshd)
6) Halt system                      15) Restore recent configuration
7) Ping host                        16) Restart PHP-FPM
8) Shell

Enter an option: █
```

Figure 3: pfSense Console Showing System Status and Interface Assignment

implemented in pfSense, including interface-based firewall policies, traffic logging, DNS/IP filtering, and network-flow monitoring.

4.2 Network Interface Configuration

The network interfaces of the pfSense firewall were configured to establish connectivity between the external network and the internal SOC laboratory segments. The interface configuration defines the network boundaries through which traffic is routed and subsequently evaluated by the firewall.

The configured interfaces are:

Interface	Device	IP Address	Network	Function
WAN	em0	192.168.241.132/24	192.168.241.0/24	External network connectivity
LAN	em1	192.168.10.1/24	192.168.10.0/24	Primary internal laboratory network
OPT1	em2	192.168.20.10/24	192.168.20.0/24	Additional internal laboratory segment

The **WAN interface (em0)** is assigned to the external-facing network and provides the upstream connectivity required by the laboratory.

The **LAN interface (em1)** is configured with 192.168.10.1/24 and functions as the gateway for the primary internal laboratory network. Systems connected to this segment use the pfSense LAN interface for network communication outside their local subnet.

The **OPT1 interface (em2)** is configured with 192.168.20.10/24 and provides a separate internal network segment. The use of an additional interface allows specific laboratory systems and traffic to be logically separated from the primary LAN network.

This interface-based design ensures that traffic entering pfSense can be associated with a specific network segment. Consequently, firewall policies, logging, and monitoring can be applied according to the interface through which traffic enters the firewall.

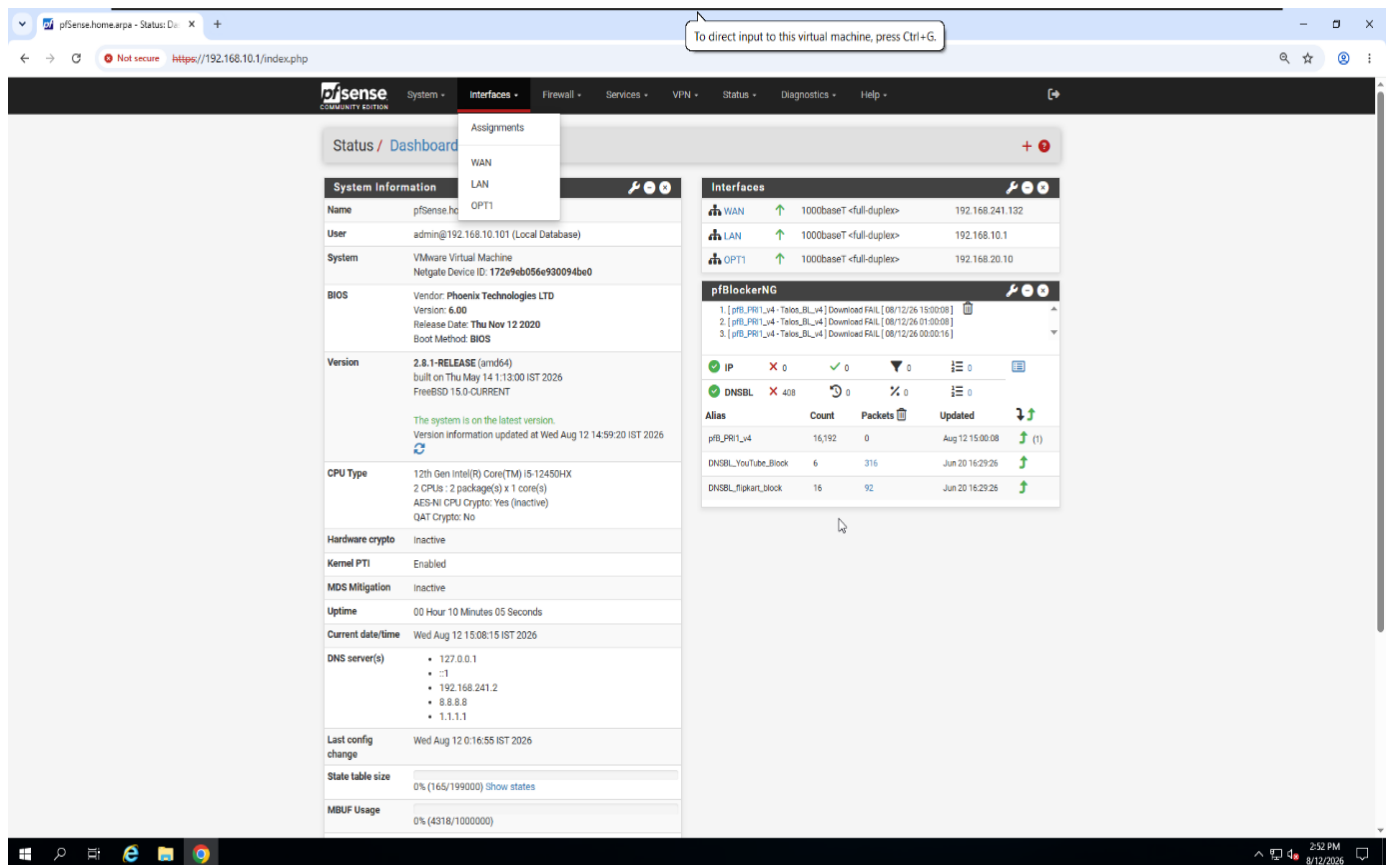


Figure 4: pfSense Dashboard Showing WAN, LAN and OPT1 Interface Configuration

The configured interface structure also provides the foundation for the later implementation of firewall rules, traffic logging, and network-flow monitoring within the SOC environment.

4.3 Firewall Traffic & Logging

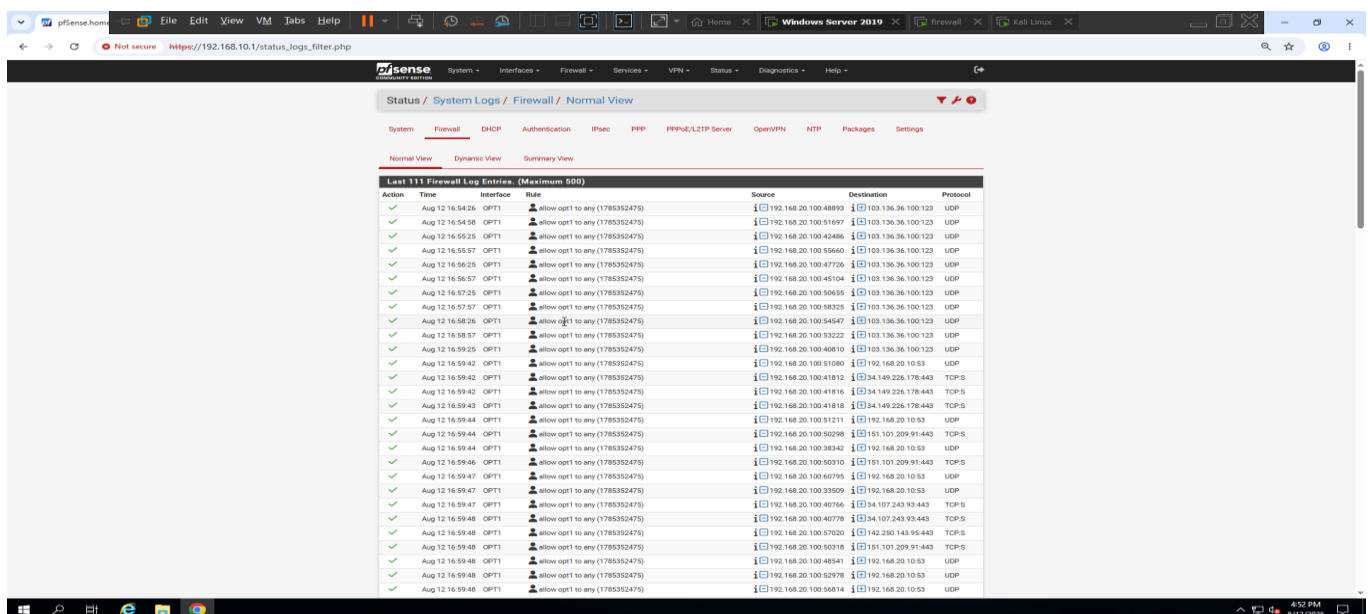
The pfSense firewall was used as a network telemetry source for the SOC environment. Firewall logging provides visibility into network connections observed at the security boundary and records important attributes of the communication, including the source address, destination address, protocol, ports, interface, and the firewall rule associated with the traffic.

The captured firewall logs show network activity originating from the OPT1 network, including traffic from the internal host 192.168.20.100. Both TCP and UDP communications are visible in the recorded events.

The observed traffic includes connections to external destinations over commonly used services, including TCP port 443, which is associated with HTTPS communication, and UDP port 123, which is commonly used for Network Time Protocol (NTP) traffic.

The firewall log also identifies the interface and rule associated with the observed traffic. This provides useful context during security investigation because an analyst can determine where the traffic originated and which firewall policy processed it.

From a SOC analysis perspective, firewall logs provide network-level evidence that can be correlated with endpoint and SIEM data. For example, an analyst can use source and destination information from firewall events to investigate whether unusual network communication occurred during a security-testing activity.



Action	Time	Interface	Rule	Source	Destination	Protocol
✓	Aug 12 16:54:26	OPT1	allow opt1 to any (785352475)	192.168.20.100:48993	103.136.36.100:123	UDP
✓	Aug 12 16:54:58	OPT1	allow opt1 to any (785352475)	192.168.20.100:51697	103.136.36.100:123	UDP
✓	Aug 12 16:55:25	OPT1	allow opt1 to any (785352475)	192.168.20.100:42486	103.136.36.100:123	UDP
✓	Aug 12 16:55:57	OPT1	allow opt1 to any (785352475)	192.168.20.100:55660	103.136.36.100:123	UDP
✓	Aug 12 16:56:25	OPT1	allow opt1 to any (785352475)	192.168.20.100:47726	103.136.36.100:123	UDP
✓	Aug 12 16:56:57	OPT1	allow opt1 to any (785352475)	192.168.20.100:40154	103.136.36.100:123	UDP
✓	Aug 12 16:57:25	OPT1	allow opt1 to any (785352475)	192.168.20.100:56609	103.136.36.100:123	UDP
✓	Aug 12 16:57:57	OPT1	allow opt1 to any (785352475)	192.168.20.100:58325	103.136.36.100:123	UDP
✓	Aug 12 16:58:26	OPT1	allow opt1 to any (785352475)	192.168.20.100:54547	103.136.36.100:123	UDP
✓	Aug 12 16:58:57	OPT1	allow opt1 to any (785352475)	192.168.20.100:53222	103.136.36.100:123	UDP
✓	Aug 12 16:59:25	OPT1	allow opt1 to any (785352475)	192.168.20.100:40810	103.136.36.100:123	UDP
✓	Aug 12 16:59:42	OPT1	allow opt1 to any (785352475)	192.168.20.100:51080	192.168.20.10:53	UDP
✓	Aug 12 16:59:42	OPT1	allow opt1 to any (785352475)	192.168.20.100:41812	34.149.226.178:443	TCP-S
✓	Aug 12 16:59:42	OPT1	allow opt1 to any (785352475)	192.168.20.100:41816	34.149.226.178:443	TCP-S
✓	Aug 12 16:59:43	OPT1	allow opt1 to any (785352475)	192.168.20.100:41818	34.149.226.178:443	TCP-S
✓	Aug 12 16:59:44	OPT1	allow opt1 to any (785352475)	192.168.20.100:51211	192.168.20.10:53	UDP
✓	Aug 12 16:59:44	OPT1	allow opt1 to any (785352475)	192.168.20.100:50298	151.101.209.91:443	TCP-S
✓	Aug 12 16:59:44	OPT1	allow opt1 to any (785352475)	192.168.20.100:38342	192.168.20.10:53	UDP
✓	Aug 12 16:59:46	OPT1	allow opt1 to any (785352475)	192.168.20.100:50310	151.101.209.91:443	TCP-S
✓	Aug 12 16:59:47	OPT1	allow opt1 to any (785352475)	192.168.20.100:60795	192.168.20.10:53	UDP
✓	Aug 12 16:59:47	OPT1	allow opt1 to any (785352475)	192.168.20.100:33509	192.168.20.10:53	UDP
✓	Aug 12 16:59:47	OPT1	allow opt1 to any (785352475)	192.168.20.100:40766	34.107.243.93:443	TCP-S
✓	Aug 12 16:59:48	OPT1	allow opt1 to any (785352475)	192.168.20.100:40778	34.107.243.93:443	TCP-S
✓	Aug 12 16:59:48	OPT1	allow opt1 to any (785352475)	192.168.20.100:57020	142.250.143.95:443	TCP-S
✓	Aug 12 16:59:48	OPT1	allow opt1 to any (785352475)	192.168.20.100:50318	151.101.209.91:443	TCP-S
✓	Aug 12 16:59:48	OPT1	allow opt1 to any (785352475)	192.168.20.100:48541	192.168.20.10:53	UDP
✓	Aug 12 16:59:48	OPT1	allow opt1 to any (785352475)	192.168.20.100:52978	192.168.20.10:53	UDP
✓	Aug 12 16:59:48	OPT1	allow opt1 to any (785352475)	192.168.20.100:56814	192.168.20.10:53	UDP

figure 5: pfSense Firewall Log Showing Observed Network Traffic

The captured logs therefore demonstrate that pfSense was actively generating network-security telemetry that could be used for monitoring and subsequent investigation within the SOC environment.

4.4 NetFlow / softflowd Network Monitoring

To provide additional visibility into network activity, NetFlow monitoring was implemented using softflowd within the pfSense environment. Unlike individual firewall log entries, flow monitoring provides summarized information about network communications and helps identify overall traffic patterns.

The softflowd statistics captured during the laboratory activity demonstrate active network-flow processing. The monitoring output records 3,419 expired flows and 3,419 exported flows, with a total of 6,735 flow records generated during the observed period.

The monitoring process received 63,494 packets through the packet-capture interface. The statistics report zero packets dropped by libpcap, indicating that the monitoring process successfully received the captured traffic during the displayed measurement period.

Protocol-level statistics provide additional information about the observed network

activity. The captured traffic includes:

- **TCP:** 38,239 packets
- **UDP:** 10,281 packets
- **ICMP:** 43 packets
- **IPv6-ICMP:** 37 packets

These statistics provide a broader view of network behavior than individual firewall events. From a SOC perspective, flow-level telemetry can assist in identifying communication patterns, dominant protocols, unusual traffic volumes, and potential anomalies that may require further investigation.

```
Flows expired: 3419 (0 forced)
Flows exported: 3419 (6735 records) in 257 packets (0 failures)
Packets received by libpcap: 63494
Packets dropped by libpcap: 0
Packets dropped by interface: 0

Expired flow statistics:  minimum      average      maximum
Flow bytes:              48           8042       15344280
Flow packets:            1           14         14330
Duration:                0.00s        6.78s      2865.75s

Expired flow reasons:
tcp = 0      tcp.rst = 422    tcp.fin = 137
udp = 2846   icmp = 14    general = 0
maxlife = 0
over 2 GiB = 0
maxflows = 0
flushed = 0

Per-protocol statistics:  Octets      Packets    Avg Life   Max Life
icmp (1):                11983      43         120.70s    380.97s
tcp (6):                 23466250   38239      21.09s     2865.75s
udp (17):                4016021    10281      3.62s      964.04s
ipv6-icmp (58):          2768       37         3.02s      8.00s
[2.8.1-RELEASE][root@pfSense.home.arpal/root: ]
```

Figure 6: softflowd NetFlow Statistics Showing Captured Network Activity

The NetFlow data therefore complements the firewall logs presented in the previous section. Firewall logs provide detailed records of individual network events, while flow statistics provide an aggregated view of network activity over the monitoring period.

4.5 pfSense to Splunk Integration

The network telemetry generated by pfSense was integrated into Splunk Enterprise to provide centralized collection and analysis of firewall-related security events. This integration is an important component of the SOC architecture because raw network activity becomes significantly more useful when it can be searched, filtered, correlated, and investigated together with events from other security sources.

The integration allows firewall activity generated by pfSense to be accessed from the Splunk search interface. Instead of relying exclusively on the pfSense console for reviewing network activity, the collected events can be examined from the centralized SIEM platform. This provides a common investigation environment for the security analyst.

4.5.1 Firewall Event Ingestion

The Splunk search results demonstrate that pfSense firewall events were successfully available within the SIEM. The events can be identified using the pfSense source and contain network-related information generated by the firewall.

The collected information provides visibility into characteristics such as:

- Event timestamp
- Source IP address
- Destination IP address
- Source and destination ports
- Network protocol
- Firewall interface
- Firewall rule information
- Network connection activity

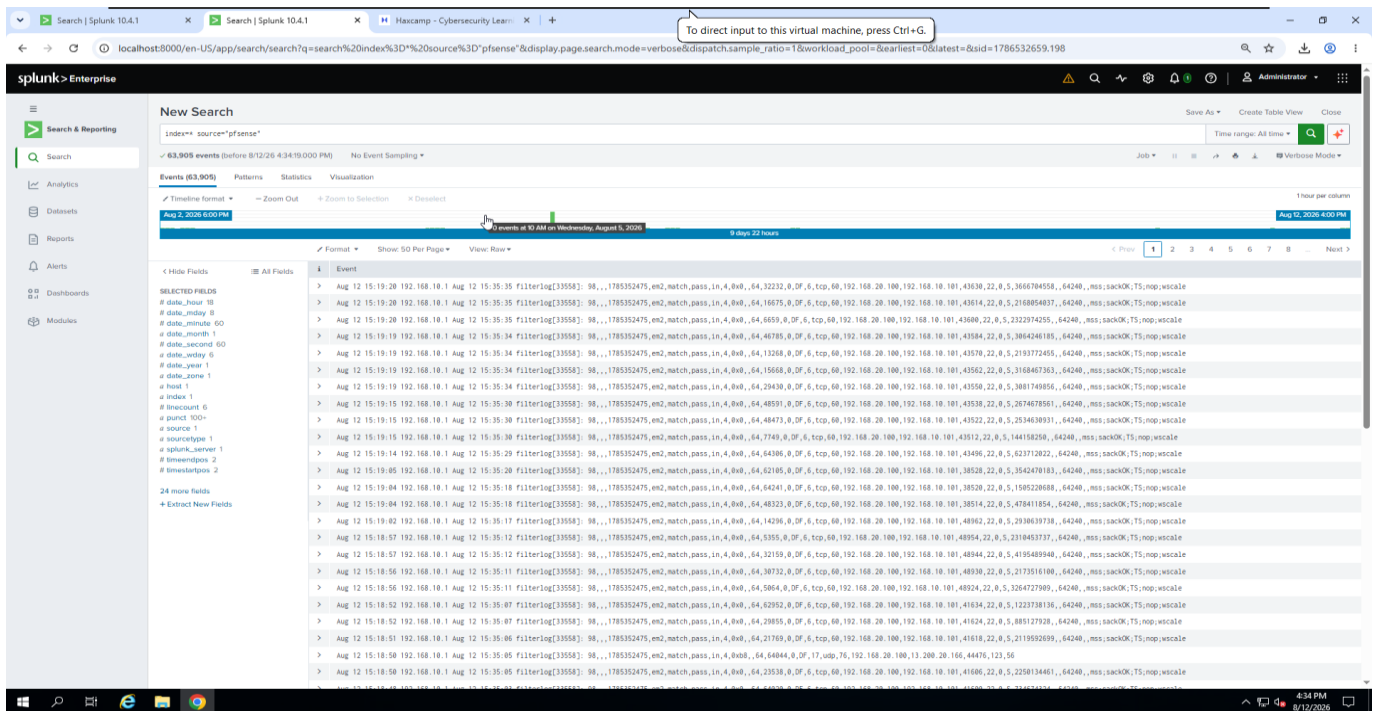


Figure 7: pfSense Firewall Events Available in Splunk

This information allows an analyst to move from a general observation of network activity to a more focused investigation of individual communication events.

4.5.2 Centralized Network Visibility

Centralizing pfSense events in Splunk provides an important advantage over analyzing firewall logs independently. A firewall is primarily responsible for enforcing network policy and recording network activity, whereas a SIEM provides tools for organizing and analyzing that information.

Within Splunk, an analyst can narrow the collected data using fields such as source address, destination address, protocol, port, and time. This makes it possible to isolate specific types of activity from a larger event set.

For example, an analyst investigating a suspicious internal host can search for all network events associated with that source address. The investigation can then identify the external destinations contacted by the host, the protocols used, and the time at which the communication occurred.

Similarly, traffic associated with a particular destination or port can be isolated to determine whether multiple internal systems communicated with the same external service.

This capability is particularly useful in a SOC environment because security investigations often begin with a small indicator and require the analyst to search a much larger collection of events.

4.5.3 Correlation with Other Security Telemetry

The primary value of integrating pfSense with Splunk is not simply storing firewall logs. The greater benefit is the ability to analyze network telemetry alongside other security data sources.

The laboratory contains both network-level and endpoint-level telemetry. pfSense provides network activity, while Windows systems generate security and authentication events. Kali Linux is used to generate controlled security-testing activity.

These different sources can provide complementary evidence during an investigation.

For example, if a controlled authentication-testing activity produces multiple failed authentication events on a Windows system, the corresponding network communication can potentially be investigated through pfSense telemetry. The analyst can examine the source system, timing, destination, and associated network activity rather than investigating the authentication events in isolation.

This creates a multi-layer investigation process:

Endpoint Event → Network Activity → SIEM Search → Event Correlation → Investigation

The ability to correlate these sources improves the analyst's understanding of what occurred and helps distinguish normal activity from activity requiring further investigation.

4.5.4 SOC Investigation Workflow

The pfSense-to-Splunk integration supports the following investigation workflow:

1. Event Generation

Network communication occurs within the laboratory environment.

2. Firewall Observation

pfSense processes the communication and generates firewall telemetry.

3. Log Collection

The generated firewall events become available to the SIEM.

4. Centralized Search

The analyst searches the pfSense event source within Splunk.

5. Event Filtering

Relevant events can be narrowed using time, source, destination, protocol, or port information.

6. Correlation

Network events can be examined alongside endpoint or authentication events.

7. Investigation

The analyst determines whether the observed activity represents normal communication, expected testing activity, or potentially suspicious behavior.

This workflow demonstrates the transition from network-device logging to SOC-level security analysis.

4.5.5 Security Monitoring Value

The integration provides several practical benefits to the SOC Home Lab.

First, it centralizes network-security telemetry so that analysts do not need to manually inspect the firewall for every investigation. Second, the large event dataset allows historical activity to be searched according to specific indicators. Third, network events can be correlated with endpoint activity to provide additional context during an investigation.

The integration also establishes a foundation for developing more advanced monitoring capabilities. Once network telemetry is available within Splunk, searches and detection logic can be developed to identify patterns such as repeated connections, unusual destinations, unexpected protocols, abnormal traffic behavior, or activity associated with controlled security testing.

Therefore, the pfSense-to-Splunk integration represents the transition from network security enforcement to centralized SOC monitoring. pfSense provides the network-level evidence, while Splunk provides the platform through which that evidence can be searched, correlated, and investigated.

5. WINDOWS SERVER AND SECURITY EVENT COLLECTION

5.1 Windows Server as a Security Telemetry Source

Windows Server was incorporated into the SOC Home Lab as an endpoint and infrastructure system from which security-relevant events could be generated and monitored. Unlike the network-level telemetry obtained from pfSense, Windows provides host-level information that describes activity occurring directly on the operating system.

This distinction is important during security investigations. Network telemetry can show that communication occurred between two systems, but endpoint telemetry can provide additional information about what happened on the affected Windows system.

The Windows environment therefore acts as a source of security events associated with user authentication, account activity, system operations, and other operating-system events. These events form an important part of the evidence available to the SOC for investigating suspicious activity.

5.2 Windows Security Event Logging

Windows records security-related activities through the Windows Event Log infrastructure. Security events can include successful authentication, failed authentication, account-related activity, privilege-related events, and other security-relevant operations.

For SOC monitoring, authentication events are particularly valuable because repeated authentication failures can indicate activities such as incorrect credentials, password-guessing attempts, or controlled brute-force testing.

One of the important events observed during the laboratory testing was **Event ID 4625**, which represents a failed logon attempt. Repeated occurrences of this event can provide evidence of abnormal authentication activity when the frequency, timing, source, and account involved are examined together.

The event data can provide investigation fields such as:

- Event ID
- Timestamp
- Account information
- Logon type
- Source network address
- Authentication information
- Failure reason
- Target system

These fields allow an analyst to move beyond simply counting failed logins and investigate the characteristics of the activity.

5.3 Security Event Collection

To make Windows security events available for centralized SOC analysis, the relevant event data was collected from the Windows environment and made available to the monitoring infrastructure.

The collection process allows security events generated on the Windows system to be examined centrally rather than requiring the analyst to manually inspect the Windows Event Viewer for every investigation.

This becomes particularly useful during controlled security testing. When an authentication-testing activity is performed against the Windows system, the resulting failed authentication events can be examined after collection to determine whether the activity generated the expected security telemetry.

The collection process therefore establishes a connection between activity occurring on the endpoint and evidence available to the SOC analyst.

5.4 Authentication Activity Monitoring

Authentication monitoring was used as one of the primary security-monitoring scenarios in the laboratory.

During normal operation, isolated authentication failures may occur because of incorrect credentials or user error. However, repeated failures occurring within a short period can require additional investigation.

For this reason, the SOC analysis does not treat a single Event ID 4625 as automatically malicious. Instead, the analyst should examine the surrounding context, including:

- Number of failures
- Time interval between failures
- Target account
- Source address
- Logon type
- Related successful authentication events
- Other network activity occurring during the same period

This contextual approach reduces the risk of treating legitimate user mistakes as security incidents.

5.5 Security Testing and Event Generation

The Windows Server environment was also used during controlled security-testing activities. The objective was to generate realistic security events and verify whether the monitoring infrastructure could capture and expose those events for analysis.

When repeated authentication attempts are generated during authorized testing, Windows records the corresponding authentication failures. These events provide a measurable security signal that can subsequently be investigated through the centralized monitoring environment.

The testing therefore serves two purposes:

1. **Generate realistic security telemetry.**
2. **Validate the ability of the SOC environment to observe and investigate that telemetry.**

This approach is more meaningful than simply configuring logging and assuming that monitoring works. The security-monitoring pipeline must be tested against known activity to determine whether the expected events are actually produced and available for investigation.

5.6 Endpoint Evidence in SOC Investigation

Windows security events provide the endpoint perspective required for multi-source investigation.

For example, a series of failed authentication events can establish that repeated login failures occurred on the Windows system. The analyst can then examine the event timestamps and source information to understand the activity and determine whether it corresponds to an authorized security test or potentially suspicious behavior.

This endpoint evidence can also be considered alongside the network activity observed elsewhere in the SOC environment. The objective is not to duplicate the network analysis but to provide an additional layer of evidence describing what occurred on the target system.

The combination of endpoint and network evidence improves investigation accuracy because the analyst is not dependent on a single telemetry source.

5.7 SOC Relevance

The Windows security-event source demonstrates an important principle of SOC operations: security monitoring requires visibility at the endpoint as well as the network level.

A firewall can identify communication patterns, while Windows security events can identify activity occurring against the operating system itself. When these sources are available to the SOC, an analyst can investigate an event using both perspectives.

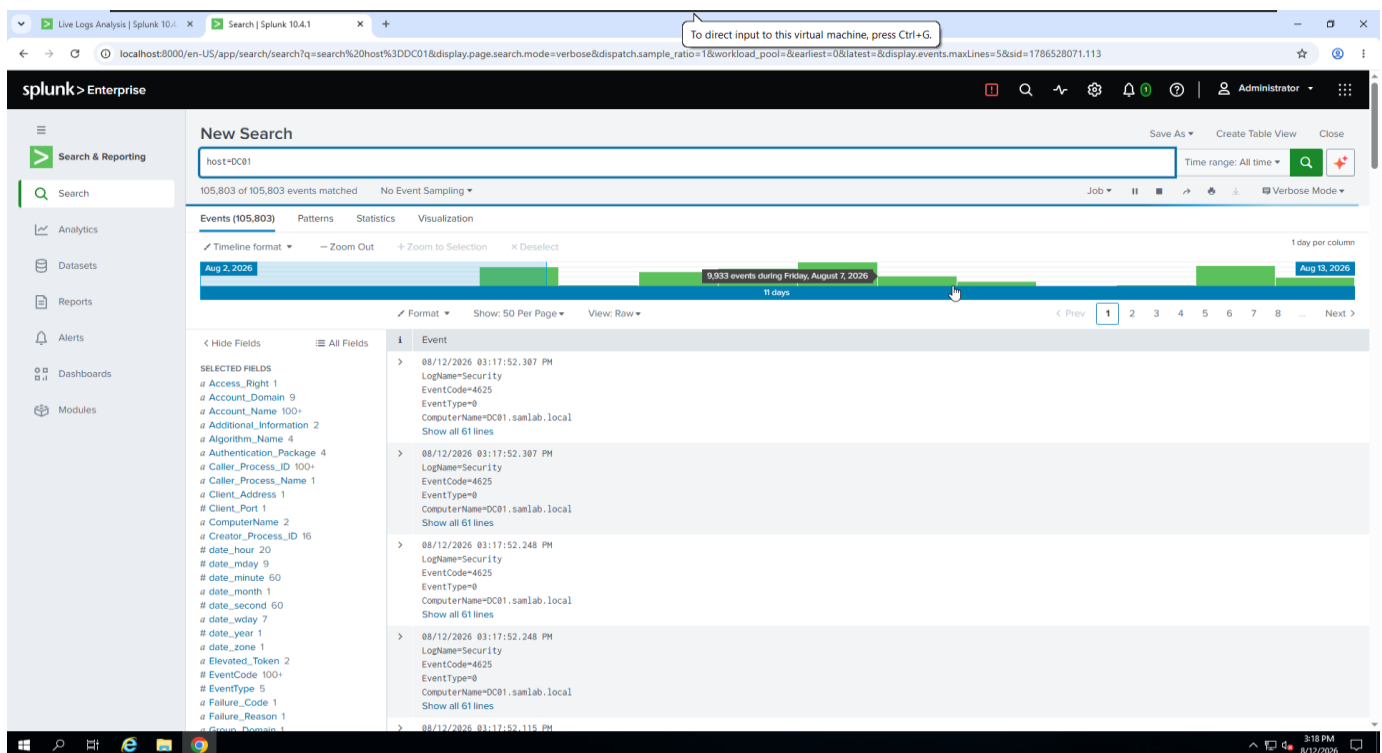


Figure 8: Windows Security Events Used as Endpoint Telemetry for SOC Monitoring

The Windows telemetry collected during this project therefore provides the foundation for the later security-monitoring and detection analysis, particularly for authentication-related activity generated during controlled testing.

6. SPLUNK SIEM IMPLEMENTATION

6.1 SIEM Implementation Overview

Splunk Enterprise was implemented as the central Security Information and Event Management (SIEM) platform within the SOC Home Lab. Its primary purpose is to provide a centralized environment in which security telemetry generated by the laboratory systems can be searched, examined, and investigated.

The implementation transforms individual security events into a centralized dataset that can be queried according to time, source, event characteristics, and other available fields. This provides the analyst with a consistent investigation interface instead of requiring separate analysis of each security component.

Splunk was configured as the primary analysis platform for the security telemetry generated during the laboratory activities. The environment includes network-related data from pfSense and endpoint-related security events from Windows systems.

6.2 Log Collection and Ingestion

Security monitoring is dependent on the availability and quality of collected telemetry. The Splunk environment was therefore configured to receive security-relevant data generated by the laboratory infrastructure.

The collected data can be distinguished according to its source. Network-security events originate from the pfSense environment, while Windows systems provide endpoint and authentication-related events.

Once the data becomes available in Splunk, the events can be searched using the Splunk Search & Reporting interface. Source-based searching allows the analyst to isolate a particular telemetry stream before performing more detailed analysis.

This approach is useful during investigation because a large SOC dataset can contain thousands of events. Filtering the dataset by source, time range, or event characteristics reduces the amount of information that must be manually examined.

6.3 Splunk Search and Event Analysis

The Splunk search interface was used to examine collected security events. Searches can be constructed to identify specific sources or event characteristics and can then be refined according to the investigation requirement.

For example, pfSense-related events can be isolated by searching for the corresponding source, allowing network-security telemetry to be examined independently from other data.

Similarly, Windows authentication events can be searched using their event identifiers. This allows the analyst to identify patterns within authentication activity rather than manually reviewing the entire Windows event stream.

The search capability provides several useful investigation functions:

- Filtering events by time
- Identifying the originating source
- Searching for specific event identifiers
- Examining individual event fields
- Counting repeated events
- Identifying source and destination information
- Narrowing large datasets to relevant activity

These capabilities form the basic analytical workflow used throughout the SOC laboratory.

6.4 Centralized Event Visibility

One of the main advantages of the implemented SIEM architecture is centralized visibility.

Without a SIEM, an analyst would need to inspect pfSense logs, Windows Event Viewer, and other sources separately. This makes investigation slower and increases the possibility of missing relationships between events.

Splunk provides a common analysis platform where different telemetry sources can be examined using a consistent search interface.

The centralized approach also makes historical investigation possible. An analyst can select an appropriate time range and examine activity that occurred before, during, or after a suspected security event.

6.5 Event-Based Investigation

The SOC Home Lab uses an event-based investigation approach. Rather than assuming that every logged event represents a security incident, events are examined in context.

A typical investigation begins with an indicator such as:

- Repeated authentication failures
- Unusual network communication
- Unexpected destination
- Abnormal traffic pattern
- Large number of similar events

The analyst can then use Splunk to determine when the activity occurred, which system generated it, how frequently it occurred, and whether related events were present.

This approach is important because individual events often lack sufficient context to determine whether activity is malicious, legitimate, or generated by authorized testing.

6.6 Security Event Correlation

The centralized SIEM environment also provides the foundation for correlating different types of telemetry.

For example, an authentication-related investigation may begin with repeated Windows security events. The analyst can examine the event timestamps and available source information and then investigate corresponding network activity during the same period.

Similarly, network activity identified through pfSense can be investigated alongside endpoint events to determine whether the observed communication corresponds to an expected system operation or a security-testing activity.

Correlation therefore allows the analyst to construct a more complete timeline of an event.

The laboratory demonstrates the following analytical relationship:

Security Event → Source Identification → Time Analysis → Related Events → Correlation → Investigation

6.7 SOC Monitoring Capability

The Splunk implementation provides the central analytical capability required by the SOC Home Lab. It enables collected telemetry to be searched and examined rather than simply stored.

The platform therefore acts as the point where information from different security sources becomes useful for investigation. Network events, endpoint events, and controlled-testing activity can be examined according to the requirements of a particular investigation.

The implementation also provides a foundation for developing more advanced SOC capabilities such as dashboards, detection searches, alerting, event correlation, and incident investigation workflows.

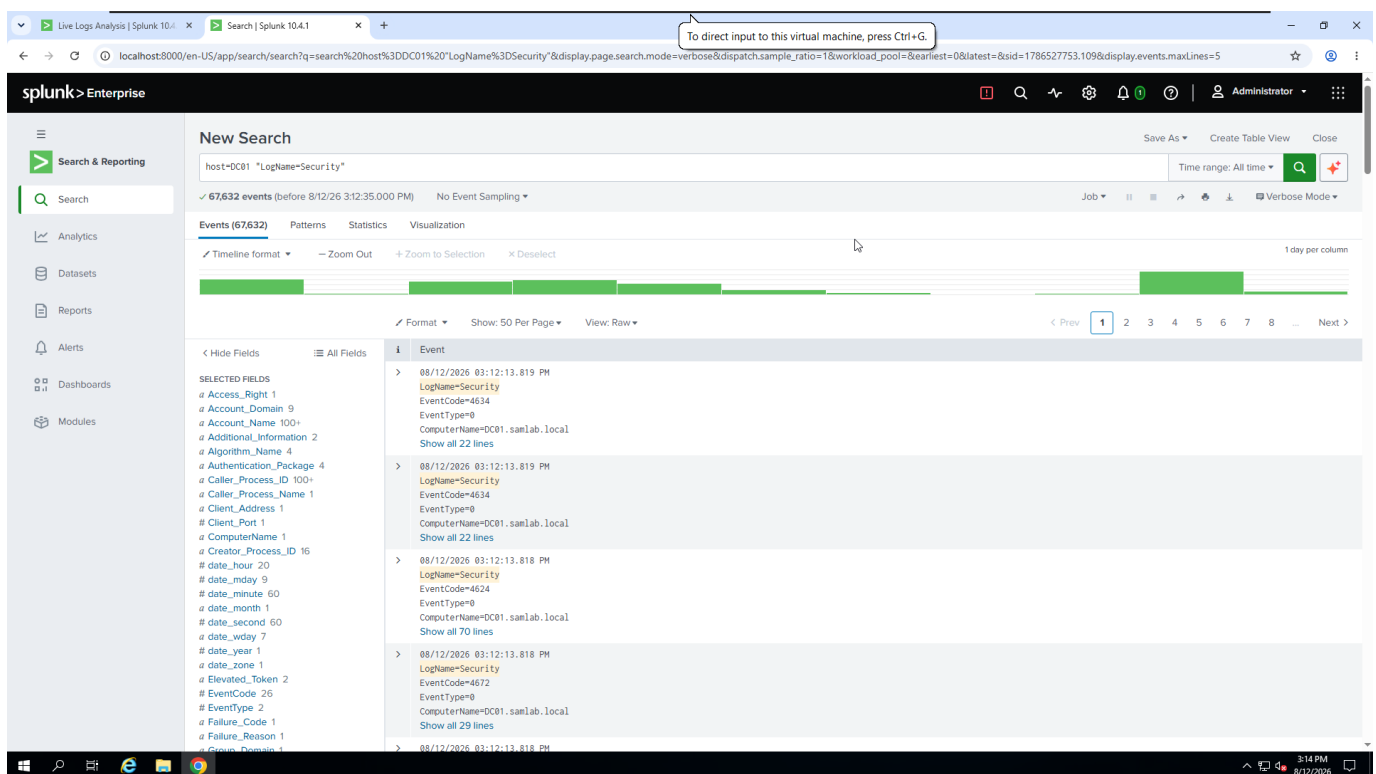


Figure 9: Splunk Enterprise SIEM Interface Showing Centralized Security Event Analysis

The effectiveness of the SIEM is subsequently evaluated through the controlled security-testing activities documented in the following chapters.

7. SECURITY MONITORING AND DETECTION

7.1 Overview

The security monitoring stage of the SOC Home Lab focuses on identifying security-relevant activity from the telemetry collected from the laboratory systems. The purpose of this stage is not simply to display logged events, but to determine whether the observed activity represents normal system behavior, authorized testing, or activity that requires further investigation.

The monitoring process uses the centralized security data available in Splunk. Events generated by Windows systems provide endpoint-level evidence, while network telemetry provides additional context regarding communication associated with the activity.

The laboratory was tested using controlled security activities so that known events could be generated and subsequently examined through the monitoring environment. This provides a practical method for evaluating whether the implemented telemetry pipeline can expose security-relevant behavior.

7.2 Authentication Activity Detection

Authentication activity was selected as one of the primary detection scenarios because repeated authentication failures can provide an indicator of attempted unauthorized access.

During the controlled testing activity, multiple failed authentication attempts were generated against the Windows environment. These attempts resulted in Windows security events corresponding to failed logon activity.

The most relevant event observed for this scenario is **Windows Event ID 4625**, which represents a failed logon.

A single failed authentication event does not automatically indicate malicious behavior. Users may enter an incorrect password, services may attempt to authenticate with outdated credentials, or configuration errors may generate isolated failures. Therefore, the detection process must consider the frequency, timing, source, target account, and surrounding activity.

Repeated occurrences within a short period provide stronger evidence that the activity should be investigated.

7.3 Splunk-Based Detection Analysis

The collected Windows security events were made available within Splunk, allowing authentication activity to be searched and analyzed centrally.

The investigation can be narrowed using the relevant Windows event identifier and other available fields. This allows the analyst to isolate failed authentication events from the larger collection of Windows telemetry.

The following characteristics are relevant during analysis:

- Event ID
- Event timestamp
- Target account
- Source address
- Logon type
- Failure information
- Frequency of occurrence

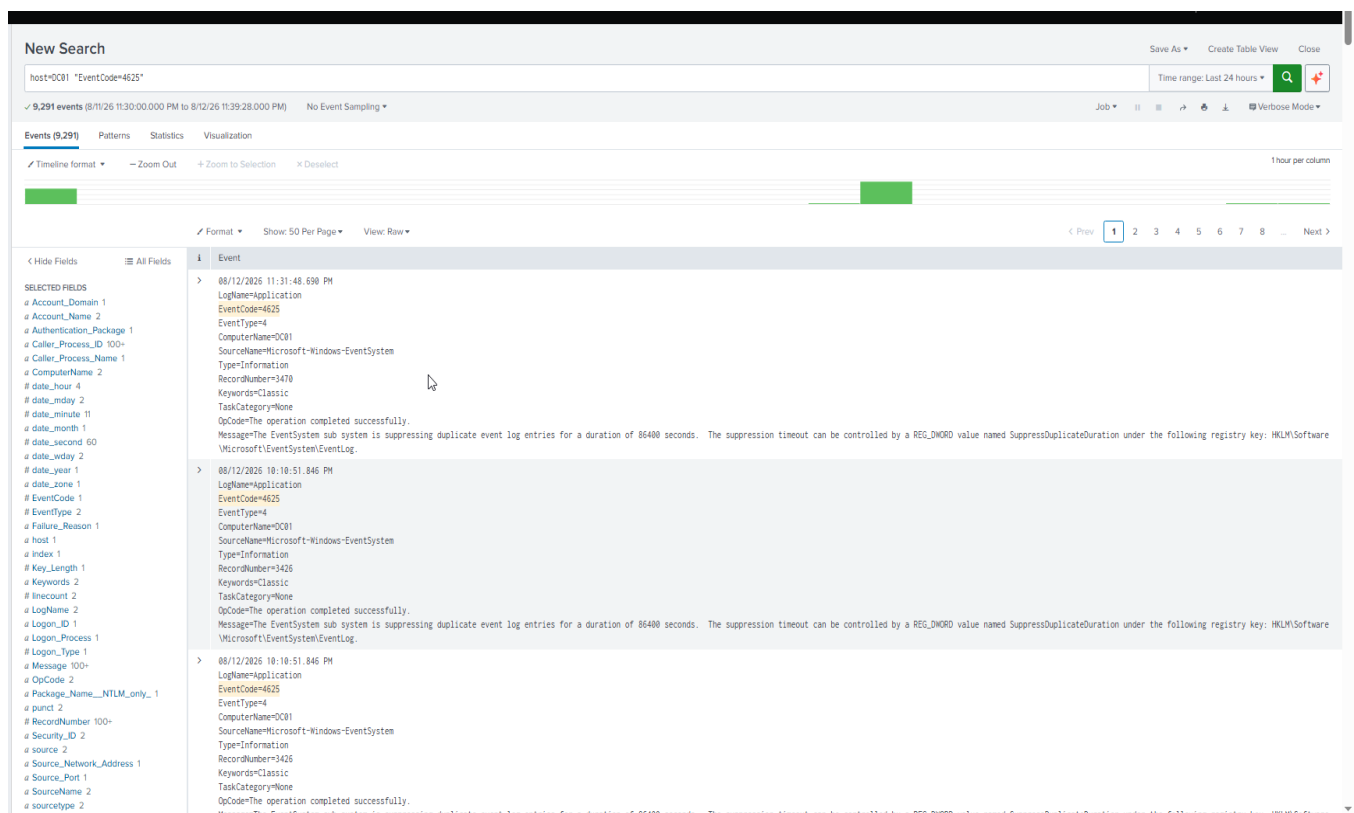


Figure 9: Splunk Analysis of Failed Authentication Events

The analyst can use these attributes to determine whether the observed authentication failures form an isolated event or a repeated pattern.

7.4 Detection of Repeated Authentication Failures

Repeated authentication failures are more significant when multiple events occur within a relatively short period and originate from the same source or target the same account.

In the laboratory scenario, the controlled security-testing activity generated repeated authentication failures. These events provide a known test case for evaluating the detection capability of the SOC environment.

The purpose of this test is to determine whether the security monitoring infrastructure can capture the events generated by the activity and make them available for investigation.

The detection process can therefore be represented as:

Controlled Authentication Activity → Windows Event 4625 → Event Collection → Splunk Search → Pattern Identification → Investigation

This workflow demonstrates how a SOC analyst can move from an observed security indicator to supporting evidence.

7.5 Event Context and False-Positive Consideration

An effective SOC should not classify every security event as an incident. Detection must be followed by contextual analysis.

For failed authentication activity, relevant questions include:

1. Was the authentication failure expected?
2. Did multiple failures occur within a short time?
3. Did the failures originate from the same source?
4. Which account was targeted?
5. Was a successful authentication observed afterward?
6. Was the activity associated with an authorized security test?
7. Was there corresponding network activity during the same period?

These questions help distinguish legitimate authentication problems from potentially suspicious activity.

In this laboratory, the activity was intentionally generated as part of controlled security testing. Therefore, the events are treated as **test-generated security telemetry**, rather than an actual unauthorized compromise.

7.6 Detection Validation

The controlled testing scenario provides a practical method for validating the SOC monitoring workflow.

The test begins with known activity and produces a predictable type of security event.

The resulting Windows events can then be located in Splunk and examined to determine whether the monitoring infrastructure captured the activity correctly.

Successful identification of the generated Event ID 4625 events demonstrates that endpoint security telemetry was available for analysis.

Further investigation can use the event timestamps and source information to establish a relationship between the generated activity and the corresponding network activity observed elsewhere in the laboratory.

This approach provides a repeatable method for testing future detection scenarios.

Additional security events can be generated under controlled conditions and evaluated using the same monitoring and investigation workflow.

7.7 SOC Detection Workflow

The implemented detection process can be summarized as follows:

Activity Generation

Controlled security activity is performed against the laboratory system.

↓

Endpoint Event Generation

The Windows operating system records the resulting security activity.

↓

Event Collection

The security event becomes available to the centralized monitoring platform.

↓

SIEM Search

Splunk is used to locate the relevant event type.

↓

Pattern Analysis

The analyst examines frequency, timing, source, account, and related activity.

↓

Contextual Investigation

The activity is compared with other available telemetry.

↓

Classification

The activity is identified as expected laboratory testing or activity requiring further investigation.

This workflow represents the practical detection cycle implemented within the SOC Home Lab.

8. CONTROLLED SECURITY TESTING WITH KALI LINUX

8.1 Purpose of Security Testing

Kali Linux was incorporated into the SOC Home Lab as the controlled security-testing platform. The purpose of the testing phase was to generate known security activity within the isolated laboratory environment and observe how that activity appeared across the implemented monitoring infrastructure.

The testing was performed against authorized laboratory systems only. The objective was not to compromise or affect external systems, but to create measurable security events that could be used to evaluate the effectiveness of the monitoring and analysis workflow.

Controlled testing is important in a SOC environment because security monitoring should be validated against known activity rather than assumed to be functional simply because logging has been configured.

8.2 Kali Linux as the Testing Platform

Kali Linux provides a collection of tools commonly used for security assessment and network analysis. Within this project, Kali was used as the source of controlled security activity against the laboratory environment.

The testing system operates as a separate component from the monitoring infrastructure. This separation allows the generated activity to be observed by the security controls without modifying the monitoring results directly.

The testing activities were designed to produce network and endpoint telemetry that could subsequently be examined through the SOC monitoring platform.

8.3 Authentication Security Testing

One of the controlled testing scenarios involved repeated authentication attempts against the authorized Windows laboratory environment.

The purpose of this activity was to simulate a password-guessing or brute-force-style authentication pattern and determine whether repeated failed authentication attempts generated identifiable security events on the target Windows system.

The testing produced failed authentication activity that was subsequently recorded by Windows security logging. These events provided the endpoint evidence used in the authentication detection scenario discussed in the previous chapter.

The testing therefore established a known relationship between:

Controlled Kali Activity → Windows Authentication Events → SIEM Telemetry

This relationship allows the SOC analyst to verify whether the expected security events are generated when a particular type of activity occurs.

8.4 Network Traffic Generation

Kali Linux was also used to generate controlled network traffic for the purpose of observing network telemetry.

Network traffic generation allows the analyst to examine how communication appears at different monitoring layers. Traffic generated by the testing system can be observed by the network security infrastructure and subsequently represented through firewall and flow-level telemetry.

The generated traffic can provide information such as:

- Source system
- Destination system
- Protocol
- Destination port
- Packet activity
- Connection frequency
- Timing of the activity

This information can be useful when validating whether network monitoring is capable of observing activity generated during security testing.

8.5 Controlled Testing Methodology

The testing process followed a controlled methodology to ensure that the activity remained within the intended laboratory scope.

Step 1 — Identify the Target

The authorized laboratory target was identified before testing began. The target system was part of the virtual SOC environment and was accessible through the configured laboratory network.

Step 2 — Generate Controlled Activity

Kali Linux was used to generate the selected authentication or network activity against the authorized target.

Step 3 — Observe Security Telemetry

The resulting activity was observed through the available monitoring sources. Network activity could be examined through the firewall and flow-monitoring components, while authentication-related activity could be observed through Windows security events.

Step 4 — Analyze the Generated Events

The collected telemetry was examined through the SIEM to determine whether the expected events were available and whether sufficient information was present for investigation.

Step 5 — Correlate Results

The timing and characteristics of the generated activity were compared with the resulting security events to establish whether the monitoring environment correctly represented the test activity.

8.6 Network-Level Visibility During Testing

Security testing also provides an opportunity to validate network-level visibility.

When Kali generates traffic toward another laboratory system, the communication passes through the configured network infrastructure according to the laboratory topology. This creates network telemetry that can be examined independently from the endpoint events produced by the target system.

This distinction is important because different monitoring sources provide different levels of information.

Network telemetry can establish that communication occurred, while endpoint telemetry can provide information about the activity that occurred on the target operating system.

The combination allows the analyst to investigate a test event from multiple perspectives without relying on a single log source.

8.7 Validation of the SOC Monitoring Pipeline

The controlled testing phase provides a practical validation mechanism for the complete monitoring pipeline.

The expected sequence is:

Kali Security Testing

↓

Network / Endpoint Activity

↓

Security Event Generation

↓

Telemetry Collection

↓

Splunk Analysis

↓

Detection and Investigation

If the expected events are visible after the controlled activity, the SOC infrastructure has demonstrated that it can capture and expose the corresponding security telemetry.

If expected events are missing, the absence becomes an important finding because it may indicate a problem with logging, collection, forwarding, indexing, or detection logic.

This makes controlled testing useful not only for demonstrating attacks but also for identifying weaknesses in the monitoring architecture itself.

8.8 Security Testing Results

The controlled testing activities successfully generated security-relevant telemetry within the laboratory environment. Authentication testing produced Windows security events associated with failed logon activity, while network testing generated observable network communication.

The resulting data could then be examined through the monitoring infrastructure and used as evidence during the security-analysis process.

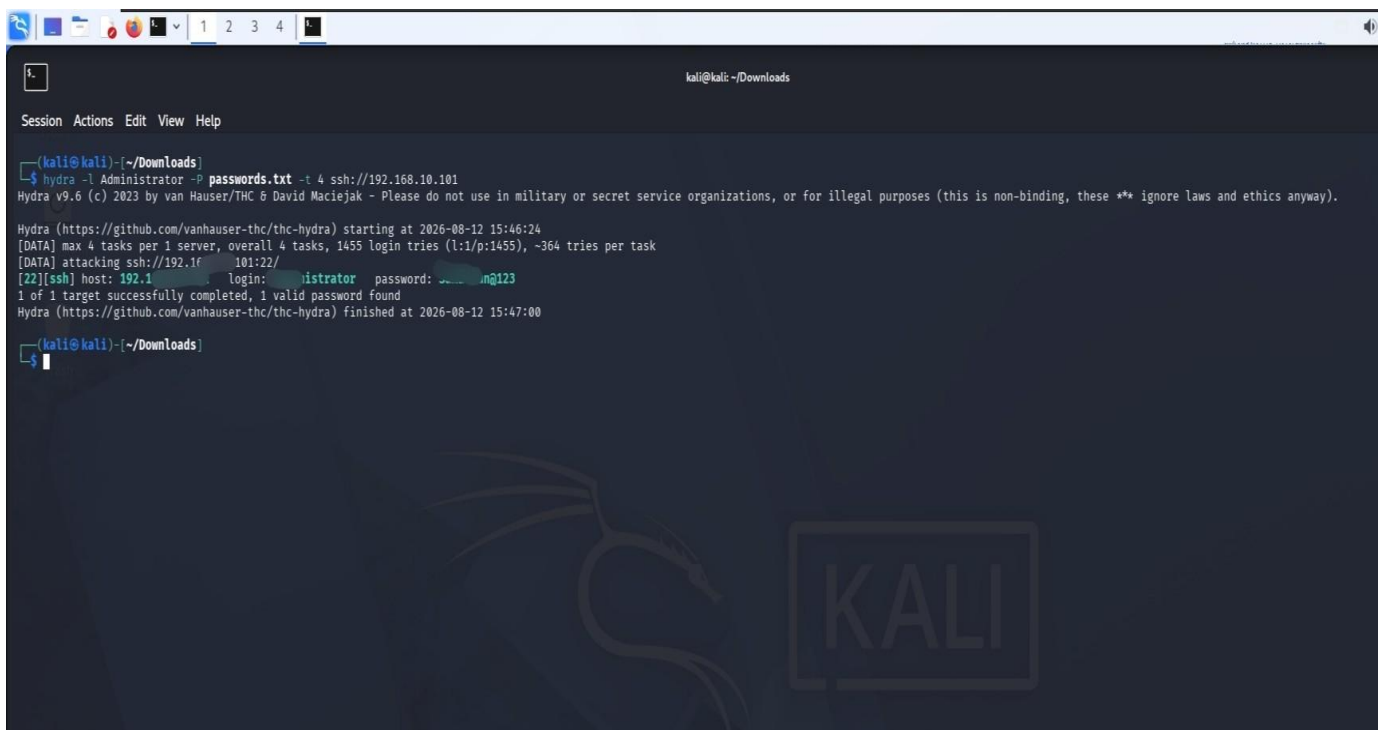
The tests demonstrate that the SOC Home Lab can be used to reproduce security scenarios in a controlled manner and subsequently analyze the resulting telemetry.

It is important to emphasize that the observed activities represent **authorized laboratory testing**. They should therefore not be interpreted as evidence of an actual external compromise or real-world intrusion.

8.9 Testing Significance

The controlled security-testing phase demonstrates the practical relationship between offensive security activity and defensive monitoring.

A security tool on its own does not constitute a complete SOC capability. The value comes from the ability to generate or observe activity, collect meaningful telemetry, identify relevant events, and investigate those events using available evidence.

A screenshot of a Kali Linux terminal window. The window title is 'kali@kali: ~/Downloads'. The terminal shows a Hydra brute-force attack on an SSH server. The command entered is 'hydra -l Administrator -P passwords.txt -t 4 ssh://192.168.10.101'. The output shows the attack starting at 2026-08-12 15:46:24, with 1455 login tries. A successful login is found for the user 'Administrator' with the password 'Administrator123'. The attack finishes at 2026-08-12 15:47:00. The terminal background features a faint Kali Linux logo and the word 'KALI' in a box.

```
kali@kali:~/Downloads
$ hydra -l Administrator -P passwords.txt -t 4 ssh://192.168.10.101
Hydra v9.6 (c) 2023 by van Hauser/THC & David Maciejak - Please do not use in military or secret service organizations, or for illegal purposes (this is non-binding, these ** ignore laws and ethics anyway).

Hydra (https://github.com/vanhauser-thc/thc-hydra) starting at 2026-08-12 15:46:24
[DATA] max 4 tasks per 1 server, overall 4 tasks, 1455 login tries (l:1/p:1455), ~364 tries per task
[DATA] attacking ssh://192.168.10.101:22/
[22][ssh] host: 192.168.10.101 login: Administrator password: Administrator123
1 of 1 target successfully completed, 1 valid password found
Hydra (https://github.com/vanhauser-thc/thc-hydra) finished at 2026-08-12 15:47:00

kali@kali:~/Downloads
```

Figure 10: Kali Linux Security Testing Activity in the Controlled SOC Environment

The Kali-based testing environment therefore provides the activity-generation component required to evaluate the defensive monitoring capabilities of the SOC Home Lab.

9. NETWORK TRAFFIC ANALYSIS

9.1 Overview

Network traffic analysis forms an important part of the SOC investigation process because network communication can provide evidence about how systems interact with internal and external destinations.

The SOC Home Lab provides network-level telemetry through firewall activity and NetFlow data. These sources can be examined to understand communication patterns, identify frequently used protocols, and establish the timing and direction of network activity.

The purpose of this analysis is not to treat every network connection as suspicious. Normal network communication generates a large amount of legitimate traffic, and effective SOC analysis requires distinguishing expected activity from patterns that warrant further investigation.

9.2 Observed Network Communication

The captured firewall telemetry shows network activity originating from systems within the laboratory environment. In particular, traffic from the OPT1 segment, including the internal host 192.168.20.100, was observed communicating with external destinations.

The firewall records provide information about the source and destination of each observed communication, together with the protocol and port involved.

Examples of observed communication include:

- UDP traffic associated with port 123
- TCP traffic associated with port 443
- Communication originating from the internal OPT1 network
- Connections toward external destination addresses

TCP port 443 represents HTTPS communication, while UDP port 123 is commonly associated with Network Time Protocol activity. These observations demonstrate that the laboratory generated both application-related and infrastructure-related network traffic.

The presence of these connections alone does not indicate malicious activity. Their significance depends on factors such as destination, frequency, timing, expected system behavior, and relationship to other security events.

9.3 Protocol Distribution

The softflowd statistics provide an aggregated view of the protocols observed during the monitoring period.

The captured statistics include:

Protocol	Packets Observed
TCP	38,239
UDP	10,281
ICMP	43
IPv6-ICMP	37

TCP represents the largest portion of the observed protocol activity, followed by UDP. ICMP and IPv6-ICMP traffic represent considerably smaller quantities within the displayed statistics.

This distribution provides a high-level understanding of the network activity generated during the observation period.

From a SOC perspective, protocol distribution can be useful as a baseline. If a future monitoring period shows a significant change in the normal protocol distribution, the difference can be investigated to determine whether it resulted from legitimate operational changes or potentially suspicious activity.

9.4 Flow-Level Analysis

The flow-monitoring data records **3,419 expired flows** and **3,419 exported flows**, with **6,735 flow records** reported by the monitoring process.

The monitoring system also recorded **63,494 packets received through libpcap**, with no packets reported as dropped by libpcap in the displayed statistics.

These values demonstrate that the monitoring system was actively processing network traffic during the observation period.

Flow-level information is useful because it summarizes communication without requiring the analyst to inspect every individual packet. This allows the analyst to identify broader patterns such as frequent communication, dominant protocols, and changes in traffic volume.

For SOC operations, this can reduce the amount of raw network information that must be manually reviewed during an initial investigation.

9.5 Source and Destination Analysis

Source and destination information is an important part of network investigation.

The firewall logs identify internal source addresses and the destinations contacted by those systems. This allows an analyst to investigate questions such as:

- Which internal host generated the communication?
- Which external destination was contacted?
- Which protocol was used?
- Which destination port was involved?
- When did the communication occur?
- Was the communication expected for that system?

For example, identifying repeated connections from one internal host to an unexpected external destination could provide a reason for further investigation.

However, destination identification alone is insufficient to classify activity as malicious. Additional context is required, including the purpose of the destination, frequency of connections, associated endpoint activity, and whether the traffic was generated intentionally during testing.

9.6 Traffic Analysis During Controlled Testing

Network telemetry becomes particularly useful when analyzed in relation to controlled security-testing activity.

The timing of generated testing activity can be compared with the network events observed by the firewall and flow-monitoring system. This provides an opportunity to determine whether the expected network activity appeared in the collected telemetry.

For example, if a controlled testing operation generates repeated connections, the analyst can examine the corresponding source address, destination, protocol, and timing within the network data.

This establishes a relationship between the activity performed by the testing system and the telemetry generated by the network-security infrastructure.

9.7 SOC Analytical Perspective

Network traffic analysis should be performed using context rather than isolated indicators.

A high volume of traffic does not automatically represent an attack. Similarly, the presence of TCP 443 or UDP 123 does not indicate malicious behavior because these protocols are commonly used by legitimate systems.

The analyst should instead look for deviations from expected behavior, such as:

- Unexpected destinations
- Unusual communication frequency
- Unexpected protocols
- Repeated connection attempts
- Unusual source systems
- Sudden changes in traffic volume
- Network activity occurring alongside suspicious endpoint events

Combining these characteristics provides a stronger basis for investigation than relying on a single network attribute.

9.8 Analysis Findings

The captured telemetry demonstrates that the SOC Home Lab successfully generated and recorded a mixture of network protocols and communication flows.

The observed data provides visibility into:

- Internal-to-external communication
- Protocol usage
- Source and destination relationships
- Network-flow volume
- Firewall-observed connections
- Traffic generated during controlled testing

The analysis also demonstrates the complementary nature of firewall logs and NetFlow data. Firewall records provide detailed information about individual observed communications, while flow statistics provide an aggregated view of network behavior.

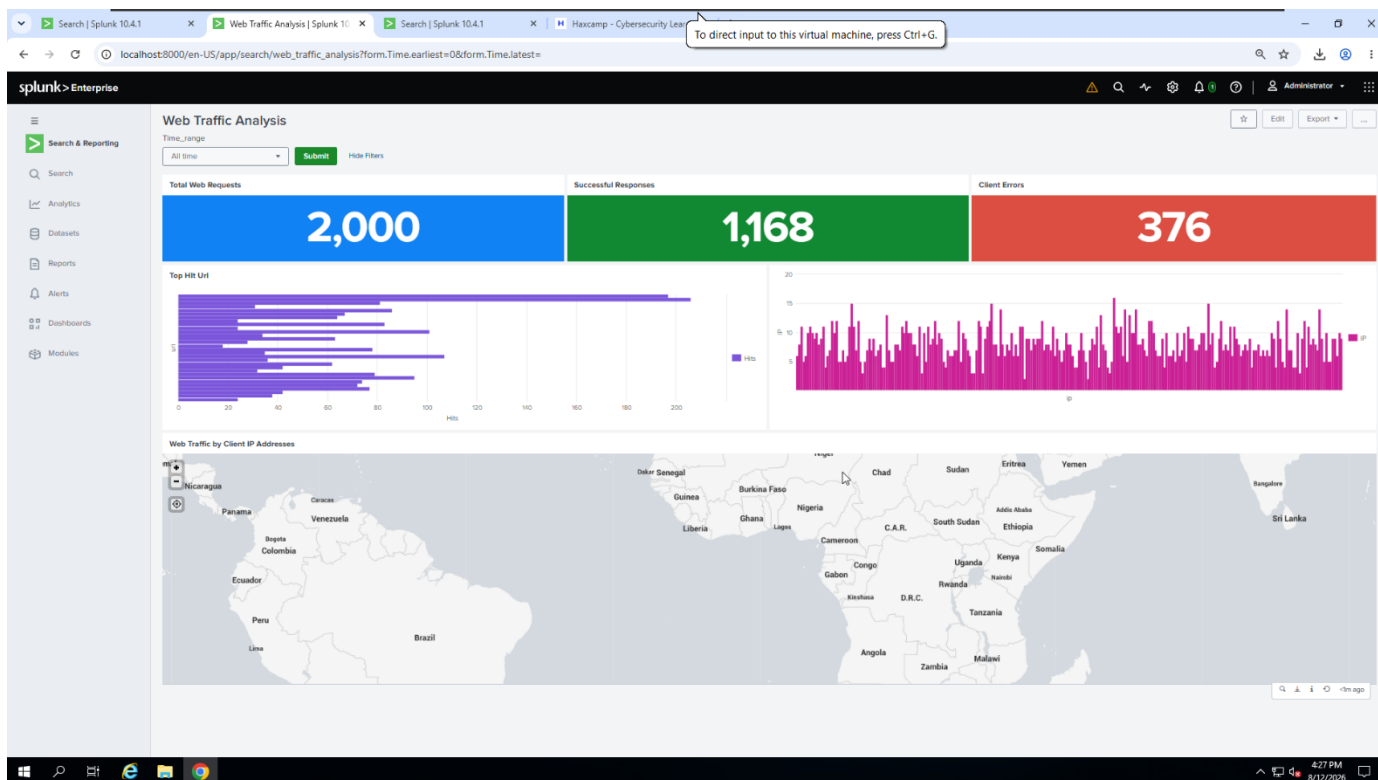


Figure 11: Splunk Web Traffic Analysis Dashboard

Together, these sources provide useful network-level evidence for SOC investigation.

10. SOC DASHBOARD AND ALERT ANALYSIS

10.1 Security Monitoring Dashboard

A dedicated security monitoring dashboard was developed in Splunk to provide a consolidated view of authentication activity within the SOC environment. The dashboard presents multiple security indicators in a visual format, allowing an analyst to quickly assess the overall state of authentication activity without manually reviewing individual events.

10.2 Authentication Activity Overview

The dashboard displays **14,650 successful logins** and **12,624 failed login attempts** within the selected dataset. These values demonstrate that the environment generated a substantial volume of authentication activity during the monitored period.

The presence of a large number of failed authentication events does not, by itself, establish that the environment was compromised. Authentication failures may occur because of incorrect credentials, automated services, configuration issues, or controlled security testing.

For this reason, the dashboard statistics should be treated as an initial indicator rather than a final security conclusion.

The dashboard also records **208 successful interactive logins**, providing a more specific view of interactive user authentication activity.

10.3 Brute-Force Alert Indicator

The dashboard reports **6 brute-force alerts**. This is a more security-relevant indicator than the total number of failed logins because it represents activity that has met the conditions defined by the implemented detection logic.

The presence of these alerts provides an analyst with a starting point for investigation.

A SOC analyst should not immediately classify an alert as a confirmed incident. Instead, the alert should be validated by examining the underlying events and determining:

- Which account was targeted
- Which source address generated the activity
- When the activity occurred
- How many attempts were generated
- Whether the attempts occurred within a short time interval
- Whether a successful login followed the failures
- Whether the activity corresponds to authorized testing

This process separates **alert generation** from **incident confirmation**.

10.4 Authentication Package Analysis

The dashboard provides a distribution of authentication packages, including **Kerberos** and **Negotiate**.

Authentication-package information provides additional context when investigating authentication events. Different authentication mechanisms can be associated with different types of Windows authentication activity, so identifying the package involved can help the analyst understand the nature of the observed event.

The dashboard visualization therefore adds another analytical dimension beyond simply counting successful and failed logins.

10.5 Success Versus Failure Analysis

The dashboard includes a comparison between successful and failed authentication activity. This allows the analyst to evaluate the relationship between authentication attempts and successful access.

A high number of failures combined with very few successful interactive logins may represent a different situation from repeated failures followed by successful access.

For security investigation, the sequence of events is therefore important.

A useful analytical workflow is:

Failed Attempts → Repeated Pattern → Detection Alert → Successful/Failed Outcome → Account Investigation

This approach provides more meaningful information than considering the total failure count alone.

10.6 Targeted Account Analysis

The dashboard identifies the accounts receiving the highest number of authentication attempts.

The displayed data includes accounts such as **DC01** and [administrator@samlab.local](#) among the highly targeted accounts.

Account targeting is an important investigation indicator because repeated authentication attempts against a specific privileged or infrastructure-related account may require additional attention.

However, the analyst must consider the role of the account and the laboratory context before determining whether the activity is suspicious.

In this environment, some of the observed activity was intentionally generated during controlled security testing. Therefore, the targeted-account information provides evidence for detection validation rather than proof of an actual unauthorized compromise.

10.7 Source Address Analysis

The dashboard also provides information about the source network addresses associated with authentication activity.

Source-address analysis helps identify where authentication attempts originated. If a large number of failures originate from a single source, the source can become an important investigation indicator.

The analyst can then examine the events associated with that address and determine whether the activity corresponds to the authorized Kali testing system or another expected laboratory component.

This allows the dashboard to act as an initial investigation interface before the analyst moves to individual events for detailed verification.

10.8 Failure Reason Analysis

The dashboard displays the reason associated with authentication failures. The observed failure reason includes:

Unknown user name or bad password

This information is useful because it provides additional context about why the authentication attempts failed.

Repeated failures caused by invalid credentials can be consistent with password-guessing activity, but they can also result from legitimate configuration problems or users entering incorrect credentials.

Consequently, failure reason should be considered together with frequency, source address, targeted account, and timing.

10.9 Authentication Timeline

The dashboard includes timelines for successful and failed authentication activity. Timeline visualization allows an analyst to identify periods in which authentication activity increases significantly.

A concentrated spike in failed authentication events can be more significant than the same number of events distributed over a long period.

Temporal analysis can therefore help identify potential bursts of activity and determine when an investigation should focus.

The timeline can also be compared against known testing periods. If an authentication spike occurs at the same time as a controlled security-testing activity, the analyst can use that information to validate the detection.

10.10 Latest Authentication Events

The dashboard provides a table of recent authentication events containing fields such as timestamp, account name, event code, logon type, authentication package, failure reason, and source network address.

This table provides a bridge between the dashboard's high-level statistics and the underlying security events.

An analyst can use the summary visualizations to identify an area of interest and then examine the latest event records for supporting evidence.

This creates a practical SOC investigation workflow:

Dashboard Overview → Identify Anomaly → Examine Relevant Events → Validate Activity → Investigate

10.11 SOC Analysis

The dashboard demonstrates how multiple authentication indicators can be presented together to support security monitoring.

Instead of relying on a single metric, the analyst can simultaneously examine:

- Overall authentication volume
- Failed authentication activity
- Successful interactive logins
- Brute-force alerts
- Targeted accounts
- Source addresses
- Failure reasons
- Authentication packages
- Activity timelines
- Individual authentication events

This combination provides a more complete view of authentication behavior.

The dashboard therefore functions as an operational monitoring interface rather than simply a graphical representation of log data. It helps the analyst identify where attention is required and provides supporting information for subsequent investigation.

10.12 Findings from Dashboard Analysis

The monitored environment demonstrated significant authentication activity, including a substantial number of failed login attempts and multiple brute-force detection alerts.

The dashboard confirms that the SOC environment was capable of identifying and presenting authentication-related security indicators in a centralized format.

However, the statistics should not be interpreted as evidence of an actual compromise without investigation. The laboratory intentionally generated security-testing activity, and the observed alerts must therefore be interpreted within that controlled context.

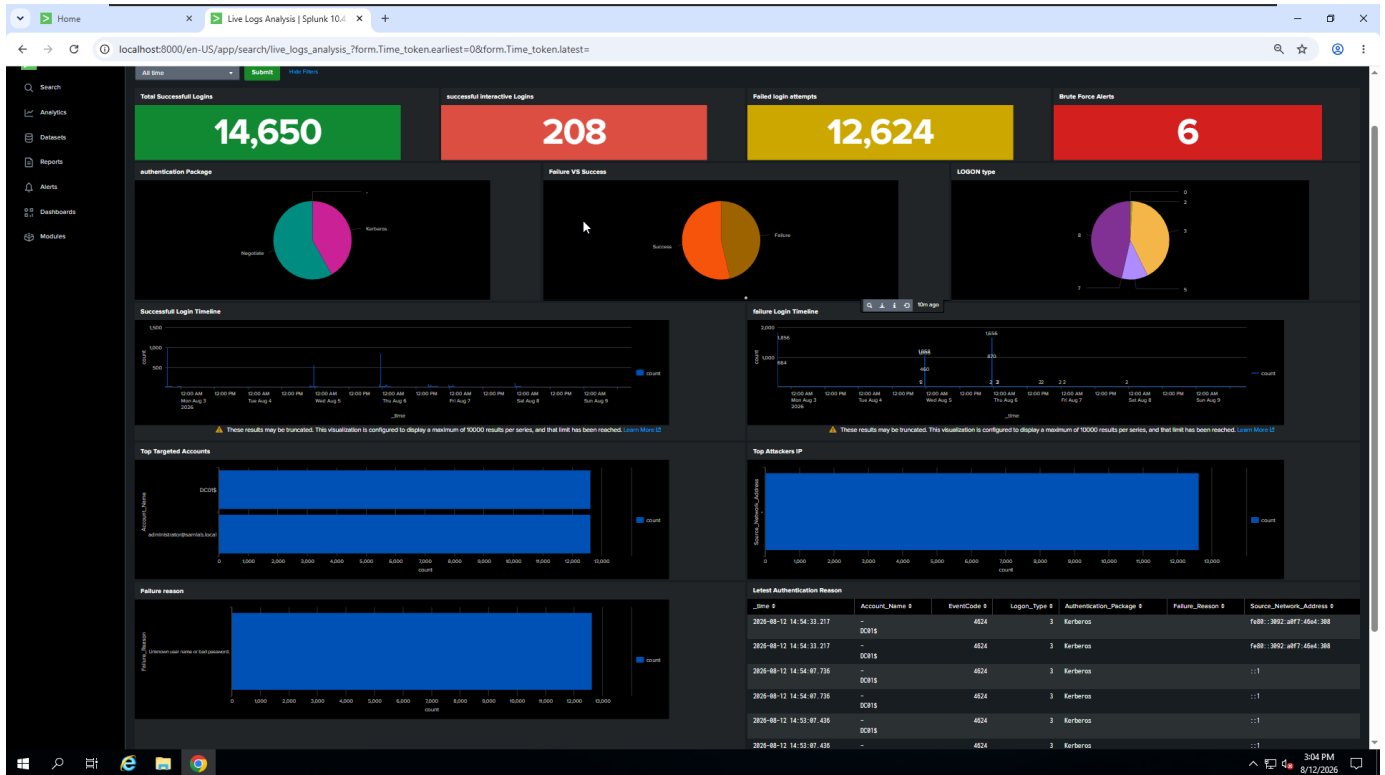


Figure 12: Splunk Live Logs Analysis Dashboard Showing Authentication Monitoring and Brute-Force Alerts

The key outcome is that the SOC monitoring environment successfully transformed raw authentication events into visual indicators, detection alerts, timelines, and investigation data that can be used by an analyst to identify and assess potentially suspicious activity.

11. FINDINGS AND SECURITY OBSERVATIONS

11.1 Authentication Monitoring and Detection

The SOC Home Lab successfully demonstrated the ability to monitor authentication activity and identify repeated failed login attempts. Windows security events, particularly **Event ID 4625**, provided endpoint-level evidence of failed authentication activity.

The Splunk dashboard further summarized this activity through failed-login counts, targeted accounts, source addresses, failure reasons, timelines, and brute-force alerts. The presence of **6 brute-force alerts** demonstrates that the monitoring environment was capable of identifying patterns that required investigation.

However, the results should be interpreted within the laboratory context. The authentication activity was intentionally generated during controlled security testing and therefore does not represent evidence of an actual unauthorized compromise.

11.2 Network and Endpoint Visibility

The project demonstrated visibility across both **network and endpoint telemetry**.

pfSense provided firewall-level network activity, while softflowd provided aggregated network-flow statistics. Windows Server supplied authentication and system security events, and Splunk provided the centralized platform for searching and visualizing the collected information.

This multi-source approach is important because a single telemetry source may not provide sufficient context for an investigation. Network data can show communication between systems, while endpoint events can provide information about activity occurring on the target system.

The combination of these sources allows the SOC analyst to investigate activity from multiple perspectives and establish a more complete understanding of an event.

11.3 Overall SOC Assessment

The implemented laboratory demonstrates the fundamental workflow of a SOC:

Security Activity → Telemetry Generation → Centralized Collection → Detection → Analysis → Investigation

The controlled testing activities successfully generated observable security events, which were subsequently available for analysis through Splunk. The project therefore validated the basic monitoring pipeline and demonstrated practical security-analysis capabilities involving authentication monitoring, firewall activity, network-flow analysis, and SIEM visualization.

The main limitation of the current implementation is that the environment is a controlled laboratory rather than a production SOC. Therefore, the results demonstrate **technical capability and detection workflow**, but should not be interpreted as production-level incident-response performance.

12. CONCLUSION AND FUTURE IMPROVEMENTS

12.1 Conclusion

The SOC Home Lab successfully demonstrated a practical security-monitoring environment using pfSense, Windows Server, Kali Linux, NetFlow/softflowd, and Splunk Enterprise. The implementation provided visibility across network and endpoint activity and allowed controlled security-testing activity to be observed through collected telemetry.

The project demonstrated the complete SOC workflow from activity generation and telemetry collection to detection, analysis, and investigation. Authentication monitoring identified repeated failed-logon activity, while firewall and network-flow data provided additional visibility into network communications. Splunk brought these data sources together for centralized searching, dashboards, and security analysis.

The controlled testing approach also validated that known security activity could generate observable events within the monitoring environment. This makes the laboratory useful for understanding how SOC analysts investigate security events rather than simply learning individual security tools in isolation.

12.2 Future Improvements

The laboratory can be further improved by implementing:

1. **Advanced Detection Rules**

Develop additional Splunk correlation searches for suspicious authentication, abnormal network activity, and other attack patterns.

2. **Automated Alerting and Response**

Configure automated alerts and response workflows so that high-confidence detections can be escalated without requiring continuous manual monitoring.

3. **Threat Intelligence Integration**

Integrate threat-intelligence feeds to compare observed IP addresses, domains, and other indicators against known malicious infrastructure.

4. **Expanded Endpoint Monitoring**

Add additional endpoint telemetry and monitoring capabilities to improve visibility into process execution, system changes, and potentially malicious behavior.

5. Incident Response Workflows

Develop structured procedures for alert triage, investigation, containment, evidence collection, and incident documentation.

6. Additional Attack Scenarios

Introduce further controlled scenarios such as suspicious network scanning, privilege-related activity, and other authorized security-testing cases to evaluate detection coverage.

12.3 Final Assessment

The completed implementation provides a functional foundation for a small-scale SOC environment and demonstrates the practical relationship between security controls, telemetry, SIEM analysis, controlled testing, and detection.

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Use the project documentation/source that corresponds to the version actually used in your lab.